

Internet Connection Types -- Guided Notes ANSWER KEY

Unit tp-2.8 | CompTIA Network+ N10-009 Obj. FC0-U71 2.8 | N10-008 FC0-U71 2.8

For instructor use / distribute after students complete the notes.

Question 1

DSL (Digital Subscriber Line) delivers internet over _____ phone lines. Download speeds are typically _____ (faster/slower) than upload because consumer DSL is _____. DSL requires a _____ at the customer premises to connect to the home network.

Answer: Copper telephone; faster; asymmetric (ADSL); DSL modem.

Question 2

Cable internet uses _____ coaxial cable, the same infrastructure used for _____ TV. It is delivered to the home via a _____ modem. Like DSL, cable is typically asymmetric -- _____ speeds are higher than _____ speeds.

Answer: Coaxial; cable; DOCSIS; download; upload.

Question 3

Fiber optic internet transmits data as _____ pulses through glass or plastic strands. It offers speeds from _____ Mbps to multiple _____ and is completely immune to _____ interference. Fiber is symmetric, meaning upload and download speeds are _____.

Answer: Light (photon); 100 Mbps; Gbps; electromagnetic (EMI); equal.

Question 4

Satellite internet works by transmitting signals to and from a _____ in orbit. Traditional geostationary satellites sit at approximately _____ km altitude, causing high _____ (100-600 ms). Low-earth orbit (LEO) satellite services like Starlink orbit at approximately 550 km, reducing latency to _____ ms.

Answer: Satellite; 35,786 km (geostationary orbit); latency; 20-40 ms.

Question 5

Cellular internet (4G LTE / 5G) uses _____ towers to deliver wireless broadband. It is the primary broadband option in _____ areas without wired infrastructure. 5G mmWave frequencies offer very high speeds but have limited _____ and cannot penetrate _____ well.

Answer: Cell; rural; range; walls (buildings).

Question 6

A _____ uses a cell phone data connection to share internet with nearby Wi-Fi devices. This is known as _____ or personal hotspot. The main limitation is that usage counts against the phone plan data _____ and battery life is _____ while hotspot is active.

Answer: Mobile hotspot (smartphone); tethering; cap (allotment); reduced.

Question 7

WiMAX is a _____ wireless broadband standard with a range of up to _____ km under ideal conditions. It was intended to provide wireless broadband to areas without cable or DSL infrastructure. It has largely been superseded by _____ and _____ in most markets.

Answer: 802.16; 50 km; 4G LTE; 5G.

Question 8 -- Real World Application

REAL WORLD: A family moves to a rural property 15 miles outside of any city. DSL and cable are not available. They need reliable internet for remote work (video calls, cloud file sharing) and school (video lessons, homework). Compare the two realistic options available -- satellite and cellular -- and recommend one based on their specific needs.

Answer: Option 1 -- Satellite (LEO like Starlink): download 50-250 Mbps, upload 10-20 Mbps, latency 20-40 ms, monthly cost higher, requires dish hardware purchase. Suitable for video calls and streaming. Option 2 -- Cellular (4G LTE): speeds vary widely by tower proximity and carrier (5-100 Mbps), latency typically 30-50 ms, monthly data caps possible, no hardware to install.

Recommendation: LEO satellite (Starlink) is better for this family because rural areas often have poor cellular coverage, and the data caps on cellular plans make heavy video call and cloud use expensive. Starlink provides consistent speeds sufficient for multiple simultaneous users and has low enough latency for video calls. The upfront hardware cost is offset by reliability.